



Dissertation on

“Bimodal Detection of Parkinson’s Disease”

Submitted in partial fulfilment of the requirements for the award of degree of

**Bachelor of Technology
in
Computer Science & Engineering**

UE20CS461A – Capstone Project Phase - 2

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CERTIFICATE

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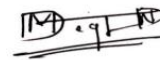
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DECLARATION

We hereby declare that the Capstone Project Phase - 2 entitled “**Bimodal Detection of Parkinson's Disease**” has been carried out by us under the guidance of Prof. Vandana M Ladwani, Assistant Professor and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology in Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester June – Nov. 2023. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ABSTRACT

Parkinson's disease is a debilitating neurological disorder which affects many people world wide. Parkinson's disease is caused by the loss of nerve cells in any part of the brain. Parkinson's disease affects the central nervous system leads to unintended or uncontrollable movements, such as shaking, stiffness and difficulty with balance and coordination. It not only affects the nervous system but also parts of the body that connects to the nervous system. As of now there is no treatment for the Parkinson's disease but there are methods to help relieve the symptoms of this disease. That is the reason detecting the disease at its early stage is very important. By considering all these symptoms and its information we are developing one software application that detects this disease at its early stage by considering its symptoms. Our main aim is to develop a machine learning model which will detect this Parkinson's disease at its early stage by using two datasets namely, spiral and vocal dataset.

In spiral dataset we are considering the spiral drawings of the the patients. In vocal dataset we are using some important features like shimmer, jitter etc. All these data are pre-processed, classified and we are using two separate models for two datasets namely, deep belief network and capsule neural network and then results are generated. Our main aim is to develop a effective tool for detecting and diagnosing the Parkinson's disease at its early stage, which can lead to improved patient outcomes and quality of life.

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CHAPTER 1

INTRODUCTION



Parkinson's disease is a neurological condition that impairs movement in sufferers. It results from the degeneration of neurons in a particular region of the brain that produce dopamine. Tremors, rigidity, slow movement, and issues with balance and coordination are all signs of Parkinson's disease. Each person experiences these symptoms differently. Other signs of this illness may include diminished sense of smell, inability to sleep soundly, anxiety, and depression, among others.

Parkinson's disease currently has no known cure, however drugs and early disease detection can help manage the symptoms and enhance quality of life for those who have the condition. The early detection of this illness enables patients to begin treatment with the goal of lessening the consequences of the symptoms. For effective disease management, our project aims to identify the illness at an early stage.

CHAPTER 2

PROBLEM STATEMENT

Parkinson's disease is a condition impacting the nervous system that results in issues with mobility, including shaking, rigidity, and challenges with maintaining balance. The objective of this project is to develop a predictive modeling solution for Parkinson's disease progression. With a focus on early detection. Timely diagnosis and adequate symptom management are imperative in dealing with Parkinson's disease. The utilization of vocal and spiral drawing information has shown effectiveness in identifying Parkinson's disease. The challenge at hand is to create a model based on machine learning that can identify Parkinson's disease with precise accuracy through vocal and spiral drawing datasets. For vocal dataset we are using Deep Belief Network and for spiral dataset we are using Capsule Neural Network. This comprehensive approach aims to significantly contribute to early detection and precise monitoring, thereby enhancing patient care and management strategies in the context of Parkinson's disease.

CHAPTER 3

LITERATURE REVIEW

3.1 “A Supervised Machine Learning Approach using Different Feature Selection Techniques on Voice Datasets for Prediction of Parkinson’s Disease”

3.1.1 Introduction

The application of supervised machine learning for Parkinson's disease prediction is discussed in this work. The rising healthcare expenditures linked to this disease, which primarily affects older individuals, are what spurred the investigation. This study focuses on leveraging speech data and various feature selection strategies to increase prediction accuracy because early detection is important for both economic and social reasons. The authors are from several academic divisions of South Korean universities, and you can get in touch with the appropriate author via phone or email.

3.1.2 Characteristics and Implementation

- The study employed voice recordings from 50 Parkinson's patients and 50 healthy controls.
- A To identify the features that were most important for classification, authors employed a range of feature selection techniques, including GA and PCA.
- Four machine learning techniques—SVM, DT, KNN, RF—were all tested for their categorization capabilities.

- The authors also assessed the performance of additional feature sets in order to determine which feature set and machine learning technique produced the best results.
- Several metrics, including AUC, accuracy, sensitivity, and specificity, were employed to evaluate each algorithm's performance.
- All of the studies were carried out with the aid of MATLAB software.

3.1.3 Features

- PD prediction using supervised machine learning is the main objective of this study. The accuracy of the predictions is improved by the authors' use of speech data and a variety of feature selection techniques.
- In the study, the effectiveness of many machine learning algorithms, including SVM, DT, KNN, and RF, is compared.
- The performance of each method is evaluated by the authors using a variety of parameters, including accuracy, sensitivity, specificity, and AUC.
- The publication provides a comprehensive discussion of the study's methodology, including feature selection techniques and machine learning algorithms.

3.1.4 Evaluation

- The sensitivity, accuracy, specificity, and AUC measures were utilised by the authors to evaluate the performance of different feature selection procedures and machine learning algorithms.
- The results showed that the GA-based feature selection technique outperformed the PCA-based technique in terms of all performance parameters.
- SVM performed the best when combined with GA-based feature selection among the machine learning methods under study.
- The researchers also compared their results to other studies in this area and found that their procedure outperformed a number of recent methods.

- The study provides a thorough examination of the data, and the conclusions are supported by tables and figures.
- The authors discuss the implications of their findings and make recommendations for further research.

3.2 “Parkinson's Disease Detection Using Voice and Spiral Drawing Dataset”

3.2.1 Introduction

This article presents a fresh approach to diagnosing Parkinson's disease using data mining techniques. The suggested method uses a combination of speech and spiral drawing datasets to forecast the severity of parkinson's disease for those who are afflicted. The report discusses the analytical procedure for these datasets, which included wavelet analysis and machine learning techniques. The results show that accuracy rates are very high in both Parkinson's disease patients and healthy individuals. This approach might help people with Parkinson's disease receive an earlier diagnosis and better care.

3.2.2 Characteristics and Implementation

- Voice dataset : made up of audio clips of the patient saying particular vowels.
- The spiral drawing dataset was produced using hand-drawn spirals on paper.
- Wavelet analysis was used to find features in the speech and spiral sketching data.
- To examine the traits that were extracted and predict the severity of parkinson's disease in those who are affected, machine learning techniques are applied.
- The best machine learning algorithm for detecting Parkinson's disease is determined by a comparison of the different methods.

3.2.3 Features

- Ingenious technique: This study suggests a novel technique for identifying PD using data mining techniques.
- Multimodal dataset: The suggested method combines speech and spiral drawing data to predict the severity of parkinson's illness in those who are affected.
- High sensitivity: The findings suggest that this method of detection has the potential to result in earlier diagnosis and better treatment outcomes for both Parkinson disease patients and healthy individuals.
- Comparison of machine learning methods: The paper examines multiple machine learning algorithms to find which algorithm has the highest accuracy rates for identifying parkinson's illness.
- Future development: The authors suggest that, in order to further increase accuracy, the model can be created as a mix of MRI or PET scans, coupled with the spiral and speech dataset.

3.2.4 Evaluation

The authors discuss the accuracy rates of various machine learning techniques used to analyse voice and spiral drawing information. They contrast their results with those of past studies on the detection of PD using comparable datasets. The authors come to the conclusion that their proposed approach has high accuracy rates for both Parkinson's disease patients and healthy individuals, demonstrating that this method of detection has the potential to enhance early diagnosis and treatment outcomes.

3.3 “Multimodal Detection of Parkinson Disease based on Vocal and Improved Spiral Test”

3.3.1 Introduction

This study focuses on the Multimodal Detection of Parkinson Disease Based on Vocal and Improved Spiral Test. The study recommends a cutting-edge technique that combines speech and image evaluations to boost the precision of PD patient recognition. The results based on the two types of datasets demonstrate a high level of PD detection accuracy. The data from the spiral and speech tests were both analysed by the authors using several machine learning models.

3.3.2 Characteristics and Implementation.

- Data on spiral drawing and speaking were supplied for the study by both PD patients and healthy controls.
- Using the Praat acoustic analysis application, 26 linear and time-frequency based features were retrieved from the voice samples.
- To minimise noise and standardise image size, a special method was applied to preprocess the spiral drawing data.
- Various machine learning models were trained on the voice and spiral test datasets to tell apart PD patients from healthy controls.
- Using a variety of criteria, including accuracy, sensitivity, specificity, and area under the curve (AUC), the authors evaluated the efficacy of each model.
- As a last suggestion, the authors suggested converting their multimodal approach into a touchbased smartphone application for quick initial PD assessments.

3.3.3 Features

- The study proposes a unique technique for diagnosing Parkinson's disease that combines speech and visual testing.
- Data on spiral drawing and speaking were supplied for the study by both PD patients and healthy controls.
- Using the Praat acoustic analysis application, 26 linear and time-frequency based features were retrieved from the voice samples.
- Using the voice and spiral test datasets, various machine learning models were created to differentiate PD patients from healthy controls.
- The authors evaluated the effectiveness of each model based on a number of factors, including accuracy, sensitivity, specificity, and area under the curve (AUC).
- The results demonstrate that the suggested multimodal technique may accurately identify PDs at a high level.
- For quick early PD assessments, the authors suggest converting their multimodal method into a touch-based smartphone application.

3.3.4 Evaluation

- The authors looked into the ability of various machine learning models to distinguish PD patients from healthy controls on the voice and spiral test datasets.
- Area under the curve (AUC), accuracy, sensitivity, and specificity were some of the metrics used to evaluate the models.
- The results demonstrate that the suggested multimodal technique may accurately identify PDs at a high level.
- The best performing model attained an accuracy of 96.25% and an AUC of 0.99 on the voice test dataset, and an accuracy of 93.75% and an AUC of 0.98 on the spiral test dataset.

- The accuracy of the multimodal approach outperformed earlier studies that only used voice or image tests for PD detection, according to the authors' comparison of their findings with those studies.
- The authors proposed to integrate their multimodal approach into a touch-based smartphone application to facilitate preliminary PD tests.
- The evaluation's overall conclusions highlight the enormous potential of the recommended multimodal strategy for accurate and useful PD identification.

3.4 “Voice-Specific Augmentations for Parkinson’s Disease Detection Using Deep Convolutional Neural Network”

3.4.1 Introduction

In order to increase the precision of Parkinson's disease identification, the paper suggests using voicespecific augmentations. The suggested method uses CNNs to extract pertinent features from voice recordings, which are then used to train a classifier to distinguish between Parkinson's disease patients and healthy listeners. According to the paper, even with a short dataset, the suggested approach achieves great accuracy in diagnosing Parkinson's disease. The possibility of the suggested approach to be applied in telemedicine applications for Parkinson's disease early detection is also highlighted in the research.

3.4.2 Characteristics and Implementation

- Based on voice-specific augmentations, the authors present a novel data augmentation technique that improves the existing voice data by introducing artificial disturbances.
- A CNN is used in the study to classify and extract features from voice samples.
- The suggested model improves classification accuracy over the baseline models by utilising the supplemented data. The authors found that their method performed better in terms of accuracy and robustness when compared to previous data augmentation strategies.

- The suggested method is implemented through pitch perturbation, white noise injection, and time stretching to produce voice-specific augmentations. The CNN architecture, which consists of many convolutional layers, pooling layers, and dense layers, is also thoroughly described by the authors.

3.4.3 Features

- The research analyses the use of various data augmentation approaches to enhance the model's accuracy and details the design of a deep convolutional neural network for this purpose.
- The study focuses on the issue of class imbalance in the dataset and offers a number of solutions to deal with it. The experimental findings presented by the authors show that the suggested way to utilise voice recordings to detect Parkinson's disease performs better than earlier methods.
- The potential of this approach to enhance patient quality of life and early Parkinson's disease diagnosis is covered in the paper's conclusion.

3.4.4 Evaluation

- The assessment findings demonstrate that the suggested strategy outperformed the conventional CNN model without voice-specific augmentations in terms of Parkinson's disease identification accuracy.
- On the two datasets, the proposed method had accuracy of 93.2% and 90.8%, respectively, But the conventional CNN model had accuracy of 90.8% and 86.6%. The study shows that the proposed method performs better in terms of accuracy than other cutting-edge methods by comparing it to them.

3.5 “A Step Towards the Automated Diagnosis of Parkinson’s Disease: Analyzing Handwriting Movements”

3.5.1 Introduction

By examining handwriting movements, the study seeks to create a system for the automated diagnosis of Parkinson's Disease (PD). The authors suggest a technique that involves gathering dynamic data generated during the writing process and processing and analysing it using machine learning techniques. According to the study's hypothesis, people with PD have distinctive writing traits that can be utilised to distinguish them from people without the disorder.

3.5.2 Characteristics and Implementation

- The article suggests an automated diagnosis approach that examines features gleaned from handwriting movements, such as pressure, speed, and curve, using machine learning techniques.
- Each participant in the study was given a set of handwriting tasks to complete, and the dataset consisted of 20 healthy people and 20 people with Parkinson's disease. A digitising tablet and a pen were used to collect the data. In order to find the most pertinent aspects for PD diagnosis, the study suggests a feature extraction and selection method.
- To distinguish between healthy people and people with Parkinson's disease (PD), a support vector machine (SVM) classifier was trained using the collected features. Using common classification criteria including accuracy, sensitivity, and specificity, the performance of the suggested technique was assessed.
- With an accuracy of 95% in classifying PD patients and healthy people, the study yields encouraging results. The suggested strategy may help with the early detection and monitoring of PD patients, which will improve treatment outcomes.

3.5.3 Features

- The study suggests a methodology for analysing handwriting samples collected from Parkinson's disease patients and healthy individuals using machine learning techniques.
- The authors stress the necessity of a Parkinson's disease early diagnosis as well as the possible advantages of an automated diagnostic system based on handwriting analysis.
- The research approach entails gathering samples of participant handwriting, extracting attributes from the samples, and using machine learning algorithms to categorise the samples as belonging to a Parkinson's disease patient or a healthy person.
- The experiment findings presented in the research show the efficiency of the suggested strategy, with a classification task accuracy rate of 97.7%.
- The proposed method may be used as an early Parkinson's disease diagnosis tool, and it may also help create a non-invasive, affordable diagnostic system, according to the scientists.

3.5.4 Evaluation

- In order to record the handwriting movements of Parkinson's disease patients and healthy controls, the study describes a data gathering procedure involving the use of a digitising tablet and specialised software. Twenty healthy controls and 27 Parkinson's disease patients provided data for the study.
- In order to extract a set of characteristics that describe the handwriting movements, the paper introduces a feature extraction method based on the Fourier series. A support vector machine (SVM) classifier is trained to differentiate between people with Parkinson's disease and healthy controls using the retrieved features. In classifying Parkinson's disease patients and healthy controls using the suggested method, the paper reports an accuracy of 84.4%.
- The use of a unique technique for analysing handwriting movements to detect Parkinson's disease, which has the potential to be a non-invasive and affordable diagnostic tool, is one of the paper's merits.

- The study's sample size, however, is somewhat limited, which restricts how far the results may be applied. It is also unknown whether the suggested strategy would be useful for diagnosing patients with more advanced stages of the disease because the study only involves participants with early-stage Parkinson's disease.

3.6 “Deep Learning Approach to Parkinson’s Disease Detection Using Voice Recordings and Convolutional Neural Network Dedicated to Image Classification”

3.6.1 Introduction

The authors suggest a cutting-edge method for detecting Parkinson's disease that makes use of speech recordings and a CNN specifically designed for picture classification. The process entails taking Melfrequency cepstral coefficients (MFCCs) out of voice recordings and feeding them into a CNN as input. The CNN is taught to categorise the MFCCs as either Parkinson's disease-free or impacted. The proposed method demonstrated the promise of deep learning and CNNs in enhancing the accuracy and objectivity of Parkinson's disease diagnosis by displaying encouraging results in Parkinson's disease detection using speech recordings.

3.6.2 Characteristics and Implementation

- Using voice recordings and a CNN, which is generally used for image classification, the article suggests a deep learning method for identifying PD.
- The effectiveness of the suggested approach in identifying PD and the deployment of CNN for voice recording analysis may be discussed in the paper.

3.6.3 Features

- The rate of occurrence of Parkinson's disease, the available diagnostic techniques, and their shortcomings are all covered in the study.

- After that, the authors go into their suggested technique, which entails taking elements from speech recordings and applying those traits to train a CNN to classify Parkinson's disease.
- The dataset utilised to train and test the CNN, as well as the experimental set-up and assessment criteria, are all thoroughly described in the study.
- The findings demonstrate that the suggested strategy outperformed conventional machine learning techniques in identifying Parkinson's disease with high accuracy. The potential of the authors' approach for increasing the accuracy of diagnosis and therapy of Parkinson's disease is covered in their conclusion.

3.6.4 Evaluation

- The results show that the proposed method got high accuracy in detecting Parkinson's disease using voice recordings, which is a promising direction for the development of non-invasive and low-cost diagnosis methods.
- However, the study has some limitations, such as the relatively small size of the dataset used, which may affect the generalization ability of the model, and the lack of comparison with other methods in the literature.

3.7 “Parkinson Disease Detection Using Various Machine Learning Algorithms”

3.7.1 Introduction

Parkinson's disease is a dangerous disorder. Its symptoms start with slowly and at the end it will go to the worst condition. As a result, many people have trouble speaking and walking. The behavioural alterations include psychological, include depression, memory issues, exhaustion, and sleep issues. The effect of this disease varies from person to person. Still many nations are living with Parkinson's disease. It affects men more than women.

3.7.2 Characteristics and implementation

- This study compares various preprocessing, extraction, and classification machine learning algorithms.
- An analysis of research publications on various techniques for diagnosing Parkinson's disease.
- Expanding the ensemble technique for improved Parkinson's disease prediction.
- The author gathered data from 60 people, including 30 with and 30 without Parkinson's disease.
- The mobile application was used to collect the data.
- To categorise the data and produce the outcome, the author used a variety of machine learning techniques.

3.7.3 Features

- The paper investigates the application of various machine learning methods, such as support vector machines, decision trees, and Random Forests, for the diagnosis of Parkinson's disease.
- Using the F1 score, precision, and accuracy to evaluate the performance.
- Extraction of shimmer and jitter, which are present in the vocal dataset and are employed for disease detection.
- Voice recordings of people with and without Parkinson's disease from a biomodel.

3.7.4 Evaluation

Parkinson's disease patients and healthy people are distinguished using a machine learning algorithm. For the classification, a variety of AI-based models are applied, which will be helpful in detecting Parkinson's illness. The best method with a good F1 score is the Random Forest method.

CHAPTER 4

PROJECT REQUIREMENT SPECIFICATION

4.1 INTRODUCTION

- Parkinson's disease is a neurological condition that impairs movement in sufferers. It results from the degeneration of neurons in a particular region of the brain that produce dopamine. Tremors, rigidity, slow movement, and issues with balance and coordination are all signs of Parkinson's disease. Each person experiences these symptoms differently. Other signs of this illness may include diminished sense of smell, inability to sleep soundly, anxiety, and depression, among others.
- Parkinson's disease currently has no known cure, however drugs and early disease detection can help manage the symptoms and enhance quality of life for those who have the condition. The early detection of this illness enables patients to begin treatment with the goal of lessening the consequences of the symptoms. For effective disease management, our project aims to identify the illness at an early stage.

4.2 Project Scope

- The scope of this project includes the collection and preprocessing of the spiral dataset and voice dataset, feature extraction and selection, and the development of a classification model for PD detection.
- The dataset will be collected from both healthy individuals and those diagnosed with PD Feature extraction techniques.

- The selected features will be combined with the spiral test data to develop a classification model using machine learning techniques such as Support Vector Machines (SVM), Random Forests, and Artificial Neural Networks (ANNs).
- The developed model will be evaluated on a separate test dataset to assess its performance in PD detection. The final outcome of the project will be a reliable and accurate system for early detection of Parkinson's disease using a combination of the spiral test and voice dataset.

4.3 Product Perspective

- The use of voice datasets and spiral tests for Parkinson's disease provides a non-invasive and cost-effective diagnostic tool for healthcare professionals to detect and monitor the progression of the disease.
- By analyzing voice patterns and fine motor control, the product can provide quantitative measurements that could assist in the diagnosis and treatment of Parkinson's disease.
- The product can be used as part of a larger diagnostic toolkit, providing an additional layer of diagnostic information that is easy to obtain, non-invasive, and cost-effective.
- The use of voice datasets and spiral tests for Parkinson's disease could lead to better outcomes for patients and more effective use of healthcare resources.

4.3.1 Product Features

- **Voice analysis :** The product would use voice analysis to measure changes in speech patterns, such as shimmer, jitter etc for voice dataset, which could indicate the presence and progression of Parkinson's disease.
- **Spiral test :** The product would use spiral drawings to measure changes in drawing patterns, such as pressure for spiral dataset, which could indicate the presence and progression of Parkinson's disease.

- **Data integration** : The product would integrate the data collected from the voice dataset and spiral test.

4.3.2 User Classes and Characteristics

- **Patients with Parkinson's disease** : The primary user class for the product would be individuals who have been diagnosed with Parkinson's disease or are at risk of developing the disease.
- **Healthcare professionals** : The secondary user class would be healthcare professionals, who would use the product to diagnose and monitor Parkinson's disease.

4.3.3 Operating Environment

- **Hardware**: The product would require no specific hardware.
- **Software**: No particular software we are using. We are using python programming language and machine learning algorithms to detect the disease.

4.3.4 General Constraints, Assumptions and Dependencies

- **Constraints** : The data in the dataset must be of good quality and must be noiseless for best results.
- **Limited resources**: Parkinson's disease detection and management require significant resources, including healthcare providers, equipment, and time. Not all healthcare facilities may have access to the necessary resources, making it difficult to provide accurate and timely diagnosis and treatment.
- **Assumptions** : We assume that we are allowed to use other libraries for processing in machine learning.
- **Dependencies** : This software/hardware requirements may include some specific software and hardware requirements such as the machine learning algorithms etc. The availability may vary with respect to the software used.

4.3.5 Risks

- Using the product in a dense activities environment, as the model might fail to recognize the events and become incapable of providing appropriate results.

4.4 Functional Requirements

- **Validity tests on inputs** : Before any inputs are processed, they should be validated to ensure they are accurate, reliable, and relevant. This could involve checking the source of the data, verifying the quality of the data, and ensuring that the data is suitable for the intended purpose.
- **Sequence of operations** : After the inputs are validated, a sequence of operations processes the inputs and generates the desired outputs. This sequence of operations will depend on the specific task.
- **Error handling and recovery** : Errors can occur at any stage of processing, so it is essential to establish a system for error handling and recovery.
- **Consequences of parameters** : The parameters used in the processing of inputs can have significant consequences for the outputs. Therefore, it is essential to consider the effect of any parameters used, such as how they affect the accuracy, speed, or reliability of the output.
- **Relationship of outputs to inputs** : Finally, it is crucial to consider the relationship between the inputs and the outputs. Parkinson's disease is a complex condition with many different symptoms, and the outputs generated by processing the inputs should be carefully evaluated to ensure they are relevant, accurate, and useful .

4.5. External Interface Requirements

4.5.1 User Interfaces :

- **GUI interface** : The screen formats and GUI styles should be designed to be clear, simple, and easy to navigate.
- **Screen layout and standard functions** : The layout of the screen should be designed in such a way that it should be simple for the user to use.

4.5.2 Hardware Requirements

- We are not using any hardware.

4.5.3 Software Requirements

- No particular software we are using. We are using python programming language and machine learning algorithms to detect the disease.

4.6 Non-Functional Requirements

4.6.1 Performance Requirements

- The model must give high accuracy as we are using two datasets.

4.6.2 Safety Requirements

- Using the product in a dense activities environment, as the model might fail to recognize the events and become incapable of providing appropriate results.

4.6.3 Security Requirements

- The data of patients must be kept securely as the confidentiality of medical details in all medical activities is a basis for patients' trust and confidence.

CHAPTER 5

HIGH LEVEL DESIGN DOCUMENT

5.1 Introduction

High level design is the architecture and structure of a particular project and it also specifies the inter relationship between the different components that are there in the project as well as the interfaces between them.

Particularly in our project i.e “**Bimodal Detection of Parkinson's Disease**”, there are different components , sub-components and user interfaces like

- **Data collection** : This component would involve collecting the dataset from various sources that are required for the efficient diagnosis of Parkinson's Disease.
- **Data Pre processing and analysis** : This would involve the processing of the dataset and cleaning that are required before the analysis of the dataset and after this it would involve the analysis part using various machine learning algorithms .
- **Feature Extraction** : This component would involve extracting the various features from the dataset that are used for the project like the spiral drawing of a person and the images that are there in the dataset of a particular person.
- **Creation of a model** : This is a sub component of the project that would create a machine learning model for the efficient diagnosis of the disease using the extracted features as the input.
- **User interfaces** : This component would involve all the components of the interfaces that helps the patients or the users to submit the images or the spiral drawings very easily. This will act as the front part of the project.

5.2 Current System

Many scientists who designed this project used only one dataset related to the patients and some used two datasets as well to detect the disease. But they failed to produce very high accuracy. Because of the medical domain, it is very important to get very high accuracy in the project. None of the project builders used image and spiral drawing dataset as the two dataset for the project.

So in our project we are combining vocal and spiral dataset and also we are working upon the accuracy of the project and trying to improve accuracy of the project.

5.3 Design Considerations

5.3.1 Design Goals

Parkinson's disease is a progressive neurological disorder that affects movement and can be difficult to diagnose in its early stages. So the primary goal of this project is to detect this disease in early stages and make this project reliable.

There are many goals related to this project like :

- **Improving the existing tools** : There are many tools discovered already for detecting Parkinson's disease, but the accuracy plays important role in medical field if it is less than it is not acceptable. So improving the technique of detecting Parkinson's disease and improving the accuracy of the project is the goal.

5.3.2 Architecture Choices

Our choice for this system would be the microservices architecture. This architecture provides the necessary flexibility to scale, maintain and update the system as needed, while keeping the services independent and focused on their specific functionality. The modularity and independence of services will enable us to update, modify, and maintain the services without impacting other services. It will

also be easier to add new functionality, such as new machine learning algorithms or datasets, to the system without affecting the existing modules.

5.4 Constraints, Assumptions and Dependencies

5.4.1 Interoperability requirements :

- 1) **Constraints** : Security related concerns may arise if patients data is shared among healthcare providers and researchers.
- 2) **Assumptions** : We assume that healthcare providers and researchers are willing to share patient data so that they can improve Parkinson's disease treatment and care.
- 3) **Dependencies** : The dependency on the availability of patient data from various sources.

5.4.2 Interface/protocol requirements :

1) Constraints :

Technological constraints: The interface or protocol requirements need to be compatible with the existing technology and available for the treatment of Parkinson's disease.

Cost constraints: The interface or protocol requirements need to be cost-effective.

2) Assumptions :

- Patients have access to the necessary technology and equipment to use the interface or protocol requirements.
- Patients have the necessary knowledge and skills to use the interface or protocol requirements properly.

3) Dependencies :

- The interface or protocol requirements are dependent on the availability of data on Parkinson's disease detection , including its symptoms and progression.

5.4.3 Data repository and distribution requirements :

1)Constraints :

- **Privacy concerns:** Personal information of patients needs to be protected, and data sharing agreements need to be there to ensure that data is not misused and lost .

2)Assumptions :

- **Data sharing:** Researchers and institutions are willing to share data related to Parkinson's disease.
- **Data access:** Researchers and clinicians have access to the necessary data to conduct their studies.

3)Dependencies :

- Adequate funding is essential for the development of repositories and the distribution of data to researchers.

5.4.4 Performance related issues :

1) Constraints :

- Parkinson's disease is a complicated illness and heterogeneous disease, which makes it difficult to predict the performance of treatments across different patient groups.

2) Assumptions :

- Early intervention in the management of Parkinson Disease is assumed to be critical to achieve better long-term outcomes.

3) Dependencies :

- Regulatory approval is required for new treatments for Parkinson Disease to be made available to patients.

5.4.5 Availability of Resources :

1) Constraints :

- Limited availability of healthcare professionals and specialized facilities in certain areas.
- Limited access to medication therapies for people with Parkinson's disease in some countries.

2) Assumptions :

- The availability of resources is assumed to be necessary for the development and implementation of effective treatments and services for people with Parkinson's disease.

3) Dependencies :

- The development of new treatments and support services for Parkinson's disease is dependent on scientific research and clinical trials.

5.4.6 Hardware or software environment :

1) Constraints :

- **Operating System Compatibility** : Any software or hardware designed for Parkinson's patients must be compatible with the operating system that is used by the users.
- **Interoperability** : Any hardware or software designed for Parkinson's patients must be interoperable with other devices

2) Assumptions :

- We assume that all patients know the language we use in our software

3) Dependencies :

- Our project is not dependent on the Hardware Environment .

5.5 High Level System Design

5.5.1 Conceptual or logical

- **Data Acquisition** : This module is responsible for collecting the voice and spiral drawing datasets from the patients.
- **Preprocessing** : This module is responsible for filtering and normalizing the datasets collected in the first module.
- **Feature Extraction** : This module is responsible for extracting the relevant features from the preprocessed datasets.
- **Machine Learning** : This module is responsible for training the machine learning model using the extracted features.
- **Model Evaluation** : This module is responsible for evaluating the trained model's accuracy and performance.
- **User Interface** : This module is responsible for providing a user interface to the doctors and users for input and output.

5.5.2 Process View:

- The preprocessing module filters and normalizes the datasets.
- The feature extraction module extracts the relevant features from the preprocessed datasets.
- The machine learning module trains the machine learning model using the extracted features.
- The model evaluation module evaluates the trained model's accuracy and performance.

- The user interface module provides a user interface to the doctors for input and output

5.5.3 Physical View:

- The user interface module runs on the doctors' devices.

5.5.4 Module View:

- The project is divided into several modules, each responsible for a specific functionality.
- Each module has its own set of requirements, design, implementation, testing, and documentation.

5.6 Design Description

5.6.1 Data Collection Module:

- This module is responsible for collecting voice data from patients.

5.6.2 Feature Extraction Module:

- This module is responsible for extracting relevant features from the collected voice data (we are doing feature extraction in vocal dataset only). Features such as pitch, jitter, and shimmer are extracted.
- The extracted features are then used as inputs to the machine learning models.

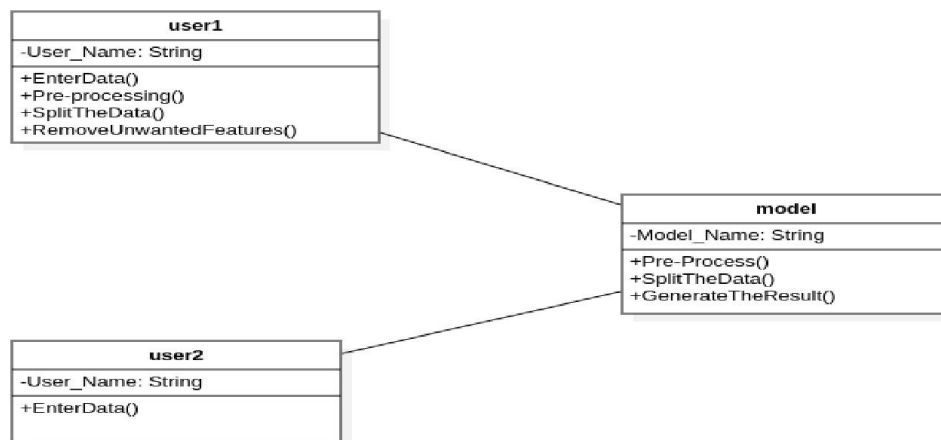
5.6.3 Machine Learning Module:

- This module is responsible for training and testing the machine learning models that detect Parkinson's disease using the extracted features.
- The module consists of several sub-modules, including data preprocessing, model training, model evaluation.
- The machine learning models are trained using voice data, and the outputs are binary classifications indicating whether a patient has Parkinson's disease or not.

5.6.4 Web Interface Module:

- This module provides a user interface for patients and doctors to interact with the system.
- The module includes several sub-modules, including user authentication, data upload, report generation. Patients can upload their voice data to the system, and doctors can access the reports generated by the machine learning models.

5.7 Master Class Diagram



5.8 Design Details

- Data Preprocessing : This stage includes cleaning of the dataset that means removing the outliers, removing the noise etc. Next is normalizing and scaling the input data for consistency.
- Cnn for spiral data : This stage includes training the spiral dataset by using the method called capsule neural network and for testing and validation also cnn is used.

-
- DBN for vocal data : Here we are using DBN model for training the vocal dataset with testing and validation.
 - Feature Integration : Here we are combining the features of both the datasets.
 - Applying the ensemble model : In this stage we are applying the ensemble model for the integrated features and getting the result.
 - Testing and validation : Here we are Reserving a portion of our dataset for testing and validation to ensure the real-world performance of our designed system.

5.8.1 Novelty :

- We are combining vocal and spiral dataset to get more accuracy.
- The novelty of this approach lies in its non-invasive, low-cost, and combination of the models.
- This approach has the potential to revolutionize the diagnosis and monitoring of Parkinson's disease, making it more accessible and effective for patients and clinicians alike.

5.8.2 Performance :

- It will depend on the quality and quantity of the data, as well as the accuracy of the models.
- The use of data augmentation, Supervised Learning techniques can help to improve the performance of the model.

CHAPTER 6

LOW LEVEL DESIGN DOCUMENT

6.1 Introduction

6.1.1 Overview

Low level design is the process of creating the detailed design specification for a particular component or software system. This is the step after high level design. Here the modules are broken down into smaller components and we will study the interaction between those components. Here it defines the flow of the data between the modules and also defines the control flow between them.

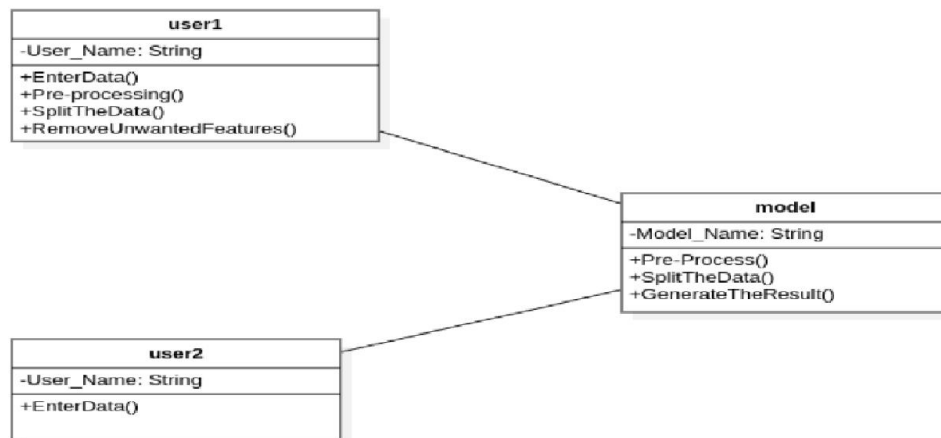
6.1.2 Purpose

- **It acts like a blueprint for the designers :** It provides a complete roadmap for the designers like how the data flow is there and the interactions between the modules etc. It mainly breaks down the whole software into smaller components.
- **It acts like a detailed version of high level design :** Actually low level design is a translation of a high level design. It gives a more detailed version of the high level design. It gives descriptions for the high level design like how each module works and all.

6.1.3 Scope

- **Specifies the data structures and algorithms :** Low level design provides the data structures that will be used in each module and it also provides the outline for the algorithms.
- **Provides optimization techniques :** low level design provides the optimization techniques for the modules.

6.2 Class Diagram

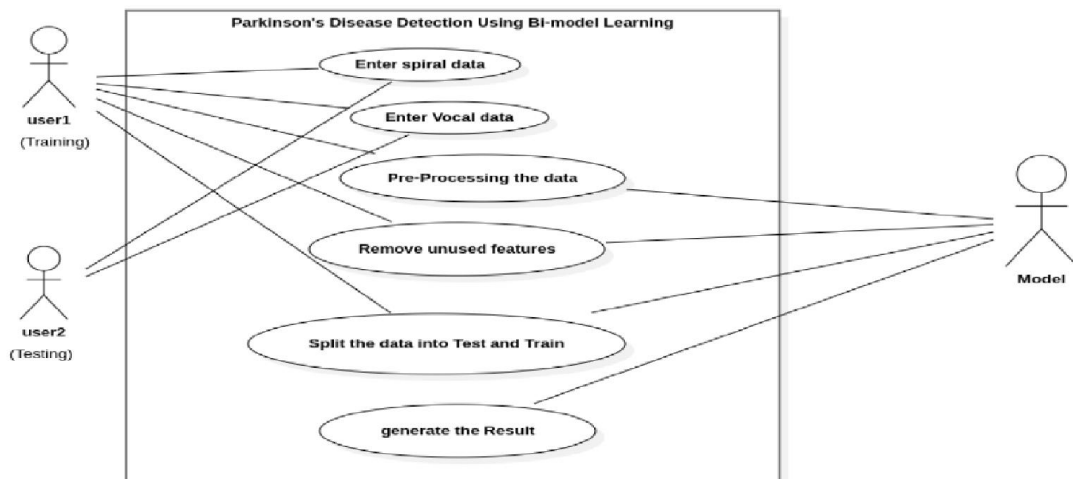


Description :

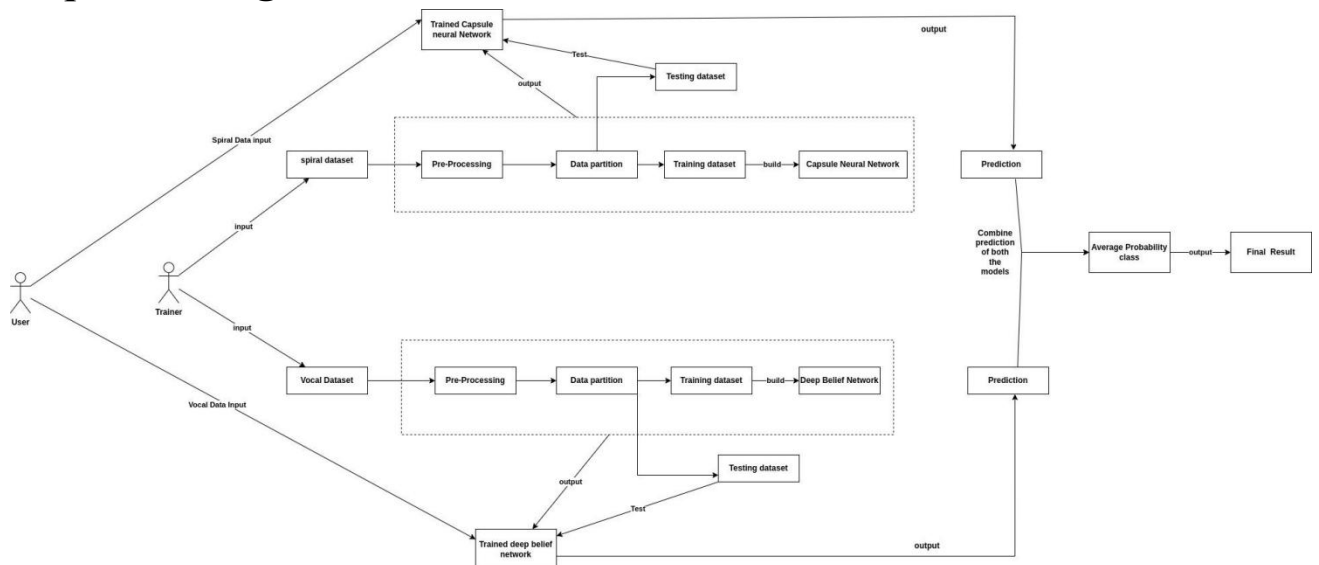
Class diagram has User 1 and user 2. Both can enter the data. User 1 acts like training modal and user 2 acts like validation modal. User1 will pre-process, split and remove unwanted data from dataset where 2 can only validate the data. The class named modal can generate the results.

6.3 Use Case Diagram

Use Case Item	Description
User1 (Training and testing)	Pre-process the data. Removes unused features, Split the data into test and train.
User2 (Testing)	Can enter the data.
Model	Generate the results .



6.4 Sequence Diagram



CHAPTER 7

SYSTEM DESIGN

7.1 Introduction

In this project we are intended to find if a person has parkinson's disease by taking spiral image input and vocal input from the user. For spiral data processing we are using a capsule neural network and for vocal data processing we are using Deep belief network

7.2 Conceptual or logical

- **Data Acquisition** : Here we collect the voice and spiral drawing datasets from the patients/ internet.
- **Preprocessing** : We check for outliers, null values etc so that we get good accuracy.
- **Feature Extraction** : This module is responsible for extracting the relevant features from the preprocessed datasets so that we don't involve unnecessary data for prediction.
- **Machine Learning** : This module is responsible for training the machine learning model using the extracted features.
- **Model Evaluation** : This module is responsible for evaluating the trained model's accuracy and performance.
- **User Interface** : This module is responsible for providing a user interface to the doctors for input and output.

7.3 Process View:

- The preprocessing module filters, segments, and normalizes the datasets.
- The feature extraction module extracts the relevant features from the preprocessed datasets.
- The machine learning module trains the machine learning model using the extracted features.

- The model evaluation module evaluates the trained model's accuracy and performance.
- The user interface module provides a user interface to the doctors for input and output

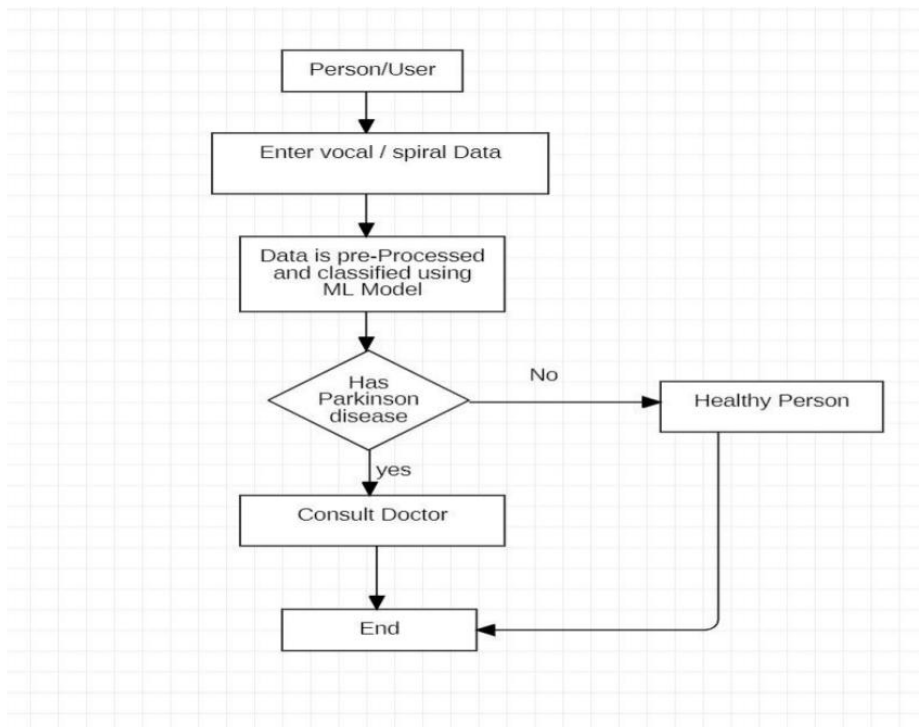
7.4 Physical View:

- The user interface module runs on the doctors' devices.

7.5 Module View:

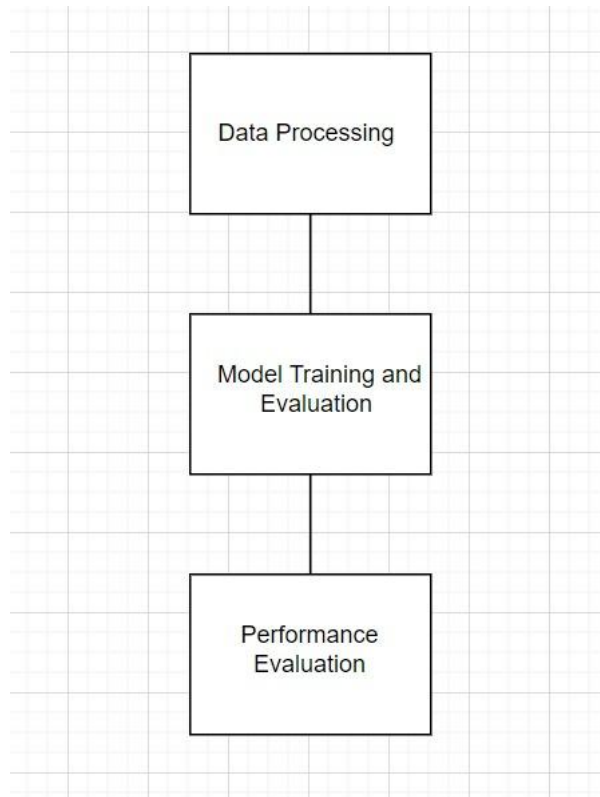
- The project is divided into several modules, each responsible for a specific functionality. Each module has its own set of requirements, design, implementation, testing, and documentation.

7.6 Block Diagram



CHAPTER 8

PROPOSED METHODOLOGY



- In both approaches, data preprocessing is a crucial first step to prepare the data for model training.
- The DBN approach focuses on deep learning with RBM layers for feature analysis, while the CNN approach utilizes Convolutional Neural Networks for binary classification.
- Both approaches involve training, validation, and testing phases to evaluate the models' performance.

CHAPTER 9

IMPLEMENTATION AND PSEUDOCODE

9.1 Implementation of DBN

9.1.1 Implementation details of DBN

- First we will import important libraries like Numpy, pandas, scikit-learn, Tensorflow, Keras etc.
- Then we will load the dataset using pandas library and select 10 most important attributes.
- Split the data into training and validation sets.
- Standardizing the data to ensure good performance.
- A Sequential model is created to build the DBN. The RBM layer with 512 hidden units is added.
- The model is compiled using Stochastic Gradient Descent (SGD) as the optimizer, binary cross-entropy as the loss function, and accuracy as the metric.
- The model is trained on the training data (X_train and y_train) for 50 epochs with a batch size of 32.
- A binary threshold of 0.5 is applied to convert the output into binary predictions.
- The trained model is saved to a file named "dbn_model_new.h5."

9.1.2 Pseudocode of DBN :

Step 1:

Initialize DBN

Add layers to the DBN

Step 2:

Pre-training of RBM layers

for each rbm_layer in dbn_layers[:-1]:

Initialize rbm_layer

Train rbm_layer on training_data

Step 3:

Fine-tuning the entire DBN

Initialize fine_tuning_parameters

for each epoch in fine_tuning_epochs:

 for each training_example in training_data:

 Forward pass through the DBN

 Backward pass and update parameters

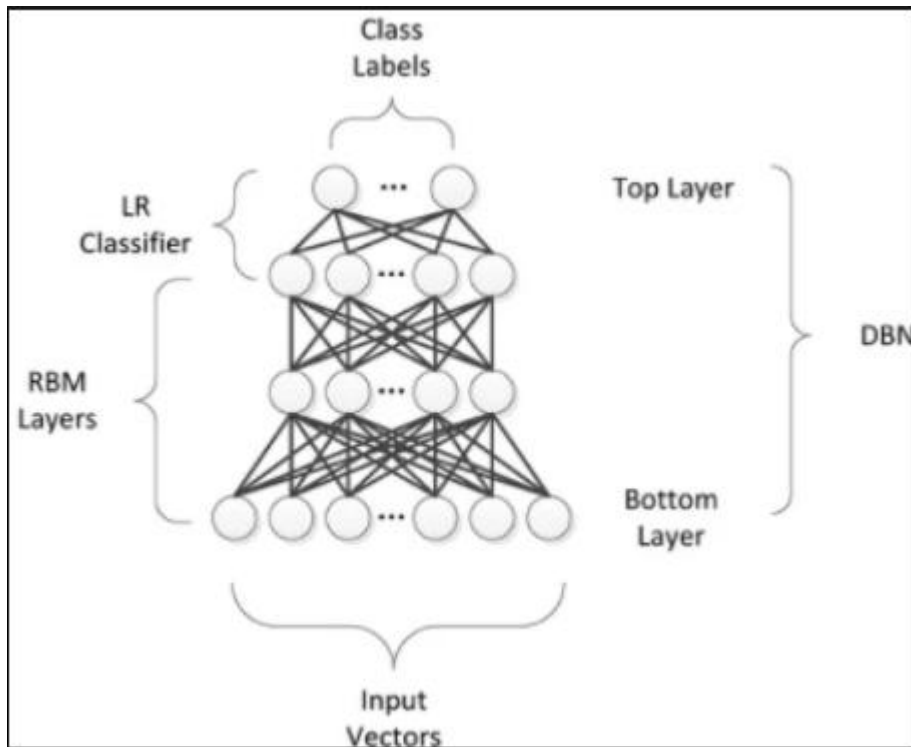
Step 4:

Make predictions

for each test_example in test_data:

 prediction = predict(test_example)

9.1.3 Architecture of DBN :



- In our project, DBN has 5 layers. one input layer with 10 neurons, 3 hidden layers with 512+256+1 neurons, 1 output layer.

9.2 Implementation of CNN

9.2.1 Implementation details of CNN

- First we will import important libraries like NumPy, Pandas, scikit-learn, TensorFlow's Keras.
- Load the dataset from a specified path and read positive , negative samples from specified folders using Pandas.
- In data preprocessing part, data is normalized using Min-Max scaling, and the dataset is split into input features and target variable , followed by an 80-20 train-test split.

- CNN model is built using TensorFlow's Keras with 1D convolutional layers, dropout, and with a sigmoid activation for binary classification.
- Model is compiled with the Adam optimizer, binary cross-entropy loss, and the accuracy metric for model evaluation.
- Training the model on trained datasets.
- Save the trained model in an H5 file format at a specified path.
- Evaluate the model on the test data and display test loss and accuracy.
- Testing model that displays the predicted labels.

9.2.2 Pseudocode of CNN :

Step 1: Initialize capsule network

```
initialize_capsule_network()
```

Step 2 : Primary capsule layer

```
for each primary_capsule in primary_capsule_layer:  
    compute_primary_capsule_activation(input_data)
```

Step 3: Capsule Layer(s)

```
for each routing_iteration in max_routing_iterations:  
    for each capsule_i in capsule_layer:  
        for each capsule_j in next_layer_capsules:  
            compute_routing_weight(capsule_i,capsule_j)  
            update_coupling_coefficients()
```

Step 4: Pose Prediction

```
for each capsule in output_layer:  
    compute_final_pose(capsule)
```

Step 5: Loss Calculation

 Compute_loss()

Step 6: Backpropagation

 backpropagate_gradients()

Step 7: Update Weights

 update_weights()

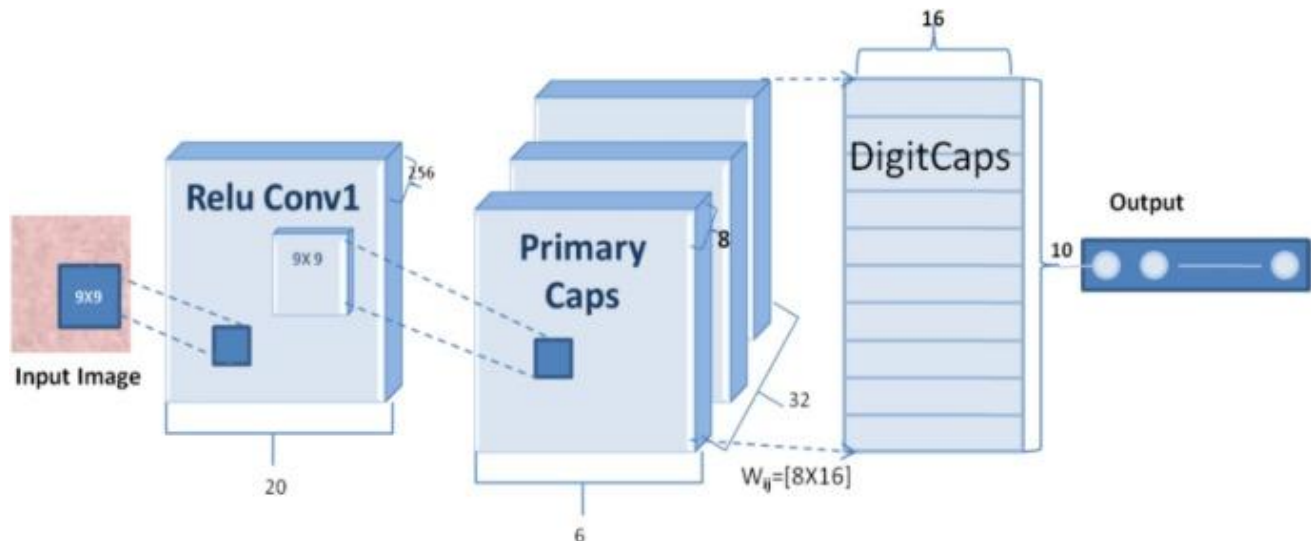
Step 8: Training Loop

```
    for each epoch in training_epochs:
        for each batch in training_data:
            forward_pass(batch)
            backward_pass()
```

Step 9: Make Predictions

```
    for each test_example in test_data:
        prediction = predict(test_example)
```

9.2.3 Architecture of CNN :



- In our project CNN has 3 layers. one input layer with 23000×4 neurons, 1 hidden layers with 256 neurons and 1 output layer.

9.3 Implementation of Ensemble Model

9.3.1 Implementation details of Ensemble Model

- Data Loading and Preprocessing
- Combines predictions from the trained CNN and DBN models.
- Calculates an average prediction for each sample.
- Converts the average predictions to binary predictions based on a threshold of 0.5.
- Evaluates the ensemble model's accuracy against the ground truth 'status' attribute from the main CSV file.
- Saves the trained CNN and DBN models to files.
- Evaluates and prints the test accuracy for both individual models and the ensemble model.

9.3.2 Pseudocode of Ensemble Model:

step 1: Initializing the List

Step 2 : Opening the csv file in read format

with open the csv file in read format:

iterating each row:

append every row's file name to list

Step 3:concatenating the file name and path

concatenating using pandas (reading csv file and join file name with path)

Step 4:Executing CNN model

Step 4.1: Initialize capsule network

initialize capsule_network()

Step 4.2 : Primary capsule layer

for each primary_capsule in primary_capsule_layer:

compute_primary_capsule_activation(input_data)

Step 4.3: Capsule Layer(s)

For each routing_iteration in max_routing_iterations:

for each capsule_i in capsule_layer:

for each capsule_j in next_layer_capsules:

compute_routing_weight(capsule_i,capsule_j)

update_coupling_coefficients()

Step 4.4: Pose Prediction

```
for each capsule in output_layer:  
    compute_final_pose(capsule)
```

Step 4.5: Loss Calculation

```
Compute_loss()
```

Step 4.6: Backpropagation

```
backpropagate_gradients()
```

Step 4.7: Update Weights

```
update_weights()
```

Step 4.8: Training Loop

```
for each epoch in training_epochs:  
    for each batch in training_data:  
        forward_pass(batch)  
        backward_pass()
```

Step 4.9: Make Predictions

```
for each test_example in test_data:  
    prediction = predict(test_example)
```

Step 5: Executing DBN model

Step 5.1: Initialize DBN

Add layers to the DBN

Step 5.2: Pre-training of RBM layers

```
for each rbm_layer in dbn_layers[:-1]:
```

Initialize rbm_layer

Train rbm_layer on training_data

Step 5.3: Fine-tuning the entire DBN

Initialize fine_tuning_parameters

for each epoch in fine_tuning_epochs:

for each training_example in training_data:

Forward pass through the DBN

Backward pass and update parameters

Step 5.4: Make predictions

for each test_example in test_data:

prediction = predict(test_example)

Step 6: Initializing ensemble list

Step 7: redimensioning and averaging two models

for every row in X_test:

redimensioning the cnn_test

redimensioning the dbn_test

taking average of two models

appending to the ensemble list

Step 8: Reading status attribute from main csv file

with open the csv file in read format:

for each row in csv:

convert status column to integer

append to list

Step 9: calculating accuracy based on status value

printing the accuracy value.

CHAPTER 10

EXPERIMENTATION RESULTS

10.1 DBN

10.1.1 Training and Testing results :

```

151/151 [=====] - 3s 20ms/step - loss: 0.0705 - accuracy: 0.9745 - val_loss: 0.1015 - val_accuracy: 0.9710
Epoch 11/50
151/151 [=====] - 3s 22ms/step - loss: 0.0706 - accuracy: 0.9763 - val_loss: 0.1126 - val_accuracy: 0.9668
Epoch 12/50
151/151 [=====] - 3s 20ms/step - loss: 0.0705 - accuracy: 0.9759 - val_loss: 0.1045 - val_accuracy: 0.9668
Epoch 13/50
...
Epoch 50/50
151/151 [=====] - 1s 9ms/step - loss: 0.0305 - accuracy: 0.9882 - val_loss: 0.1331 - val_accuracy: 0.9701
38/38 [=====] - 0s 4ms/step
Validation Accuracy: 0.9701244813278008

```

10.1.2 Validation results :

```

[0] ✓ 1.3s
.. 2/2 [=====] - 0s 11ms/step
Actual 'status' values:
[0 1 0 0 0 0 0 0 0 1 0 0 1 0 1 0 0 0 0 1 0 0 0 1 0 1 0 1 0 1 1 0 1 1
 0 1 0 0 0 1 1 1 1 0 1 0 1 0 1 1 0 0 0 1 0 0]
Predicted 'status' values:
[0 1 0 0 1 0 0 0 0 1 0 0 1 0 1 0 0 0 0 1 0 0 0 1 0 0 1 0 1 0 1 0 1 1 0 1 1
 1 1 0 1 0 1 1 1 1 0 1 0 1 0 1 1 0 0 0 1 0 0]
Accuracy: 0.9491525423728814
Confusion Matrix:
[[33  3]
 [ 0 23]]

```

10.2 CNN

```
# Calculate and print accuracy using inverted predictions
accuracy = accuracy_score(actual_values, inverted_predictions)
print(f'\nAccuracy: {accuracy * 100:.2f}%')
```

```
1/1 [=====] - 1s 1s/step
File: P_26060002.csv, Actual: 1, Predicted: 1
File: P_23100002.csv, Actual: 1, Predicted: 1
File: C_0007.csv, Actual: 0, Predicted: 0
File: P_26060001.csv, Actual: 1, Predicted: 1
File: C_0002.csv, Actual: 0, Predicted: 0
File: C_0013.csv, Actual: 0, Predicted: 0
File: C_0005.csv, Actual: 0, Predicted: 1
File: P_12060001.csv, Actual: 1, Predicted: 1
File: P_16100004.csv, Actual: 1, Predicted: 0
File: C_0014.csv, Actual: 0, Predicted: 1
File: P_26060003.csv, Actual: 1, Predicted: 0
File: C_0010.csv, Actual: 0, Predicted: 1
File: P_27110001.csv, Actual: 1, Predicted: 1
File: C_0003.csv, Actual: 0, Predicted: 1
File: C_0004.csv, Actual: 0, Predicted: 0
File: P_12060002.csv, Actual: 1, Predicted: 1
File: P_16100003.csv, Actual: 1, Predicted: 1
File: P_30100001.csv, Actual: 1, Predicted: 0
File: C_0006.csv, Actual: 0, Predicted: 0
File: P_23100003.csv, Actual: 1, Predicted: 1
File: P_30100002.csv, Actual: 1, Predicted: 1
File: C_0012.csv, Actual: 0, Predicted: 1
File: C_0009.csv, Actual: 0, Predicted: 0
File: C_0011.csv, Actual: 0, Predicted: 0
File: C_0015.csv, Actual: 0, Predicted: 1
File: C_0008.csv, Actual: 0, Predicted: 0
File: P_26060006.csv, Actual: 1, Predicted: 1
File: P_27110003.csv, Actual: 1, Predicted: 1
File: C_0001.csv, Actual: 0, Predicted: 0
File: P_26060007.csv, Actual: 1, Predicted: 1

Accuracy: 70.00%
```



10.3 Ensemble Model

```

1/1 [=====] - 1s 541ms/step
1/1 [=====] - 0s 62ms/step
Size of separate_predictions: (30, 22992, 256, 1)
Size of y_validation_pred: (30, 1)
Actual: 1, Predicted: [1.]
Actual: 1, Predicted: [1.]
Actual: 0, Predicted: [0.]
Actual: 1, Predicted: [1.]
Actual: 0, Predicted: [0.]
Actual: 0, Predicted: [0.]
Actual: 0, Predicted: 1
Actual: 1, Predicted: [1.]
Actual: 1, Predicted: 0
Actual: 0, Predicted: 1
Actual: 1, Predicted: 0
Actual: 0, Predicted: 1
Actual: 1, Predicted: [1.]
Actual: 0, Predicted: 1
Actual: 0, Predicted: [0.]
Actual: 1, Predicted: [1.]
Actual: 1, Predicted: [1.]
Actual: 1, Predicted: 0
Actual: 0, Predicted: [0.]
Actual: 1, Predicted: [1.]
Actual: 1, Predicted: [1.]
Actual: 0, Predicted: 1
Actual: 0, Predicted: [0.]
Actual: 0, Predicted: [0.]
Actual: 0, Predicted: 1
Actual: 0, Predicted: [0.]
Actual: 1, Predicted: [1.]
Actual: 1, Predicted: [1.]
Actual: 0, Predicted: [0.]
Actual: 1, Predicted: [1.]
[1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 1]
[1 1 0 1 0 0 1 1 0 1 0 1 0 0 1 1 1 0 1 1 0 0 0 0 0 0 1 1 0 1]
Accuracy: 70.00%

```

CHAPTER 11

CONCLUSION OF CAPSTONE PROJECT PHASE - 2

- Parkinson's Disease is a neurological disorder that affects many people and it leads to death. So the main objective is to build a model so that it can detect the disease in the early stage and to help the patients to cure it.
- For this we used two models and we got 97% accuracy in DBN model and 70% accuracy in CNN model.
- The main challenge we faced here was to extract the features from the raw dataset. For that we used wrapper method as the solution.
- From this, we learnt how the model works and how it trains and tests the data with the good result.
- We combined results of both the models and achieved accuracy of 70%.
- Patients can easily use this website for detection of the disease with ease. They just have to upload the voice recording and spiral drawing.
- This project can be further improved by using the advanced AI tools for the prediction of the disease.

REFERENCES/BIBLIOGRAPHY

- [1] S. Aich, H. -C. Kim, K. younga, K. L. Hui, A. A. Al-Absi and M. Sain, **"A Supervised Machine Learning Approach using Different Feature Selection Techniques on Voice Datasets for Prediction of Parkinson's Disease,"** 2019 21st International Conference on Advanced Communication Technology (ICACT), PyeongChang, Korea (South), 2019, pp. 1116-1121, doi: 10.23919/ICACT.2019.8701961.
- [2] K. M. M. Rao, M. S. N. Reddy, V. R. Teja, P. Krishnan, D. J. Aravindhar and M. Sambath, **"Parkinson's Disease Detection Using Voice and Spiral Drawing Dataset,"** 2020 Third International Conference on Smart Systems and Inventive Technology (ICSSIT), Tirunelveli, India, 2020, pp. 787-791, doi: 10.1109/ICSSIT48917.2020.9214276.
- [3] H. N. Pham et al., **"Multimodal Detection of Parkinson Disease based on Vocal and Improved Spiral Test,"** 2019 International Conference on System Science and Engineering (ICSSE), Dong Hoi, Vietnam, 2019, pp. 279-284, doi: 10.1109/ICSSE.2019.8823309.
- [4] M. Hireš, M. Gazda, L. Vavrek and P. Drotár, **"Voice-Specific Augmentations for Parkinson's Disease Detection Using Deep Convolutional Neural Network,"** 2022 IEEE 20th Jubilee World Symposium on Applied Machine Intelligence and Informatics (SAMI), Poprad, Slovakia, 2022, pp. 000213-000218, doi: 10.1109/SAMI54271.2022.9780856.
- [5] C. R. Pereira et al., **"A Step Towards the Automated Diagnosis of Parkinson's Disease: Analyzing Handwriting Movements,"** 2015 IEEE 28th International Symposium on Computer-Based Medical Systems, Sao Carlos, Brazil, 2015, pp. 171-176, doi: 10.1109/CBMS.2015.34.

[6] M. Wodzinski, A. Skalski, D. Hemmerling, J. R. Orozco-Arroyave and E. Nöth, **"Deep Learning Approach to Parkinson's Disease Detection Using Voice Recordings and Convolutional Neural Network Dedicated to Image Classification,"** 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Berlin, Germany, 2019, pp. 717-720, doi: 10.1109/EMBC.2019.8856972.

[7] K. S., A. P. and S. R., **"Parkinson Disease Detection Using Various Machine Learning Algorithms,"** 2022 International Conference on Advanced Computing Technologies and Applications(ICACTA),Coimbatore, India, 2022, pp.1-6, doi: 10.1109/ICACTA54488.2022.9752925.

APPENDIX A

DEFINITIONS, ACRONYMS AND ABBREVIATIONS

- DBN- Deep Belief Network
- CNN-Convolution Neural network
- PD-Parkinson's Disease

Bimodal Detection of Parkinson's Disease

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Abstract — Parkinson's disease is a debilitating neurological disorder which affects many people world wide. Parkinson's disease is caused by the loss of nerve cells in any part of the brain. It is a major brain disorder that affects men more than women. Parkinson's disease affects the central nervous system and leads to unintended or uncontrollable movements, such as shaking, stiffness and difficulty with balance and coordination. It not only affects the nervous system but also parts of the body that connect to the nervous system. As of now there is no treatment for Parkinson's disease but there are methods to help relieve the symptoms of this disease. That is the reason detecting the disease at its early stage is very important. By considering all these symptoms and its information we are developing one software application that detects this disease at its early stage by considering its symptoms. Our main aim is to develop a machine learning model which will detect Parkinson's disease at its early stage by using two datasets namely, spiral and vocal dataset. In the spiral dataset we are considering the spiral drawings of the patients. After collecting we are going to analyze the pen movements while drawing the spiral structure. In the vocal dataset we are collecting the patient's voice such as sustained vowel phonation, reading a sentence and spontaneous speech. All these data are preprocessed , classified and we are using two separate models for two datasets namely, deep belief network and capsule neural network and then results are generated. Our main aim is to develop a non-invasive, effective tool for detecting and diagnosing Parkinson's disease at its early stage, which can lead to improved patient outcomes and quality of life.

Keywords - Capsule Neural Network (CNN) , Deep Belief Network (DBN) , neurological disorder

I. INTRODUCTION

Parkinson's disease is a neurological condition that impairs movement in sufferers. It results from the degeneration of neurons in a particular region of the brain that produce dopamine. Tremors, rigidity, slow movement, and issues with balance and coordination are all signs of Parkinson's disease in the brain that produce dopamine. Each person experiences these symptoms differently. Other

signs of this illness may include diminished sense of smell, inability to sleep soundly, anxiety, and depression, among others Parkinson's disease currently has no known cure, however drugs and early disease detection can help manage the symptoms and enhance quality of life for those who have the condition. The early detection of this illness enables patients to begin treatment with the goal of lessening the consequences of the symptoms. For effective disease management, our project aims to identify the illness at an early stage.

II. RELATED WORK

Several past research works have contributed to the field of Parkinson's disease detection, providing valuable insights and benchmark results.

In [1], S. Aich et al. compared various models, including decision trees, support vector machines, and Dirichlet process mixtures. The Dirichlet process demonstrated the best classification performance, achieving an accuracy of 87.7% compared to other models.

Another research work[1] focused on feature selection methods and machine learning-based approaches for Parkinson's disease diagnosis. They employed mutual information-based feature selection and support vector machines, achieving an impressive accuracy of 92.75%.

K. M. M. Rao et al.[2] explored the use of different classifier algorithms, such as Naïve Bayes, Support Vector Machine (SVM), Multilayer Perceptron, and decision trees. One author used SVM and K-fold cross-validation, achieving an accuracy of 85% in identifying Parkinson's disease based on a heterogeneous acoustic database.

In a separate investigation[3], spiral and vocal datasets from the UCI repository were utilized. H. N. Pham et al. applied k-means clustering for vocal dataset feature extraction. For modeling, they employed methods like k-nearest neighbors, random forests, and SVM. In the case of the spiral dataset, they utilized SS_t, DST, and STCP tests with logistic regression, k-nearest neighbors, and adaptive boosting, respectively. Combining these approaches using an ensemble with majority voting, they achieved high accuracy rates of approximately 95% for the spiral dataset and 95.89% for the vocal dataset.

Another study focused on analyzing handwritten spiral drawings[5]. C. R. Pereira et al. extracted numeric features and employed Naïve Bayes, optimum path Forest, and SVM

with RBF kernel for classification. They conducted 10-fold cross-validation, computing measures such as recall, precision, F1 score, and accuracy. The evaluation revealed promising results.

Motivated by the aforementioned past works, our research explores different models, including deep belief networks and capsule neural networks, for Parkinson's disease detection. We compare their performance with various machine learning classifiers to gain further insights and advancements in the field.

By summarizing these related works, we establish the foundation for our research and highlight the need for further exploration and improvement in Parkinson's disease detection methodologies.

III. MATERIALS

A. Data

The Parkinson's Disease dataset consists of a vocal and spiral dataset of people who are affected by Parkinson's Disease and people who are healthy.

1) *Vocal Data Set*: The Parkinson's disease vocal dataset, encompassing 6024 voice recordings, meticulously curated 3012 instances each for individuals with and without Parkinson's disease (PD), serves as a rich repository of biomedical voice measurements. These 23 attributes, representing diverse voice measures, collectively compose a comprehensive overview of vocal characteristics. Each row in the dataset corresponds to one of the 195 voice recordings from individuals, with the "status" column crucially designating 0 for healthy subjects and 1 for those diagnosed with PD. The primary objective of this dataset is to facilitate the discrimination between healthy and PD-affected individuals based on the distinctive features encapsulated within the various voice measurements.

In its ASCII CSV format, this dataset is a valuable resource for research, with each row encapsulating a unique voice recording instance. By encapsulating a wide spectrum of vocal parameters and meticulously categorizing individuals based on their health status, this dataset not only contributes to the understanding of the vocal manifestations associated with PD but also presents a targeted tool for developing machine learning models aimed at effective disease discrimination.

TABLE I.

TIME - FREQUENCY- BASED FEATURES
EXTRACTED FROM SPEECH SAMPLES

Features	Group
MDVP:F0(Hz) MDVP:Fhi(Hz) MDVP:Flo(Hz)	Frequency Parameters
MDVP:Jitter(%) MDVP:Jitter(Abs) MDVP:RAP MDVP:PPQ Jitter:DDP	Pulse Parameters

MDVP:Shimmer MDVP:Shimmer(dB) Shimmer:APQ3 Shimmer:APQ5 MDVP:APQ Shimmer:DDA	Amplitude Parameters
NHR	Voicing Parameters
HNHR	Pitch Parameters
RPDE DFA	Harmonicity Parameters
spread1 spread2 D2 PPE	Other Parameters
status	Status Indicator

2) *Spiral Image Datasets*: For data collection utilizing a graphics tablet, three alternative test kinds have been developed.

The first one is the Static Spiral Test (SST), which has been shown to be useful in clinical research for a number of purposes, including tremor assessment and Parkinson's disease diagnosis. For data collection utilizing a graphics tablet, three alternative test kinds have been developed. The Static Spiral Test (SST) is the first and is widely used in clinical research for a number of purposes, including tremor assessment, Parkinson's disease (PD) diagnosis, and motor function assessment.

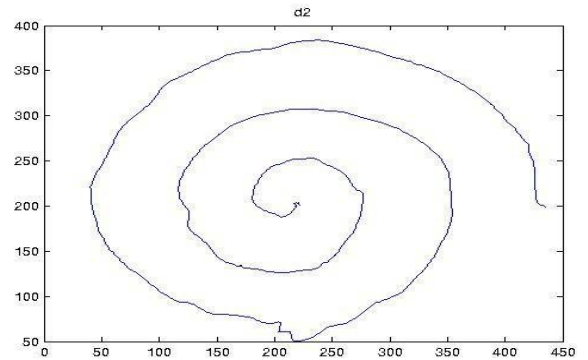


Fig. 1. Controlled

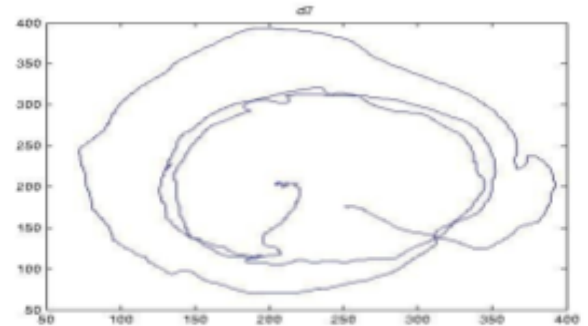


Fig. 2. With Parkinson

The following CSV variables are used to delimit the Spiral Dataset: X, Y, Z, Pressure, GripAngle, Timestamp, and Test ID.

Test ID

0: Static Spiral Test (Draw the provided spiral design)

1: Dynamic Spiral Test (The spiral design will blink after a predetermined length of time, so subjects must keep drawing.)

2: Circular Motion Test (Subjects draw circles around the red point).

A. Training Datasets:

Vocal datasets: Employing a wrapper method, we intricately navigated through the array of 23 attributes, discerningly selecting a subset of 10 attributes that demonstrated the utmost significance in influencing the classification outcomes. This meticulous attribute curation process stands as a pivotal component in refining the classification model, ensuring that the chosen attributes contribute optimally to the intricate task of distinguishing between distinct classes within the dataset.

Spiral Datasets: In the pursuit of refining classification accuracy, a discerning analysis of seven selected attributes revealed a strategic identification of the top-performing quintet. These five attributes, meticulously chosen for their paramount contributions, stand poised as instrumental factors in enhancing the precision of the classification process.

PROPOSED WORK

In our study, an ensemble model was employed for Parkinson's disease detection. The spiral dataset was processed using a capsule neural network, while the vocal dataset was analyzed using a deep belief network.

In our study, two models are used i.e unimodal and bimodal. Unimodal means a single type of data. For spiral dataset, the modality used was capsule neural network and for vocal dataset, the modality used was deep belief network. Bimodal means two types of data.

For this, two types of features were merged and the modality used was the combination of CNN and DBN.

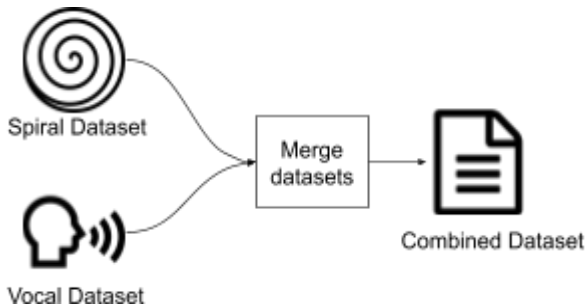


Fig. 3. Data Merging

A) Capsule neural networks

The capsule neural network (CNN) employed in this study is designed to discern patterns within the spiral dataset for binary classification. The architecture comprises multiple one-dimensional capsule layers, each followed by dropout layers to mitigate overfitting. Rectified Linear Unit (ReLU) activation functions introduce non-linearity. The global average pooling layer reduces spatial

dimensions, and a dense layer with a sigmoid activation function performs binary classification. The model is compiled using the Adam optimizer and binary cross-entropy loss function.



Fig. 4. Capsule Neural Network Unimodal

A) Capsule Operation:

$$\text{Conv}(X, W) = X * W$$

It is applied to the input data

- X with the filter (kernel)
- W, where denotes the capsule neurological operation

B) Dropout

$\text{Dropout}(X)$ = Randomly set a fraction of input elements to 0 during training

C) Global Average Pooling

$$\text{GlobalAvgPool}(X) = \frac{1}{N} \sum_{i=1}^N X_i$$

calculates the average over spatial dimensions, reducing the output to a single value for each feature map.

D) Sigmoid Activation

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

E) Stochastic Gradient Descent (SGD)

SGD Update rule:

$$W_{t+1} = W_t - \eta \nabla J(W_t)$$

B) Deep Belief Network

A DBN model is constructed using Restricted Boltzmann Machines (RBMs). The DBN includes RBM layers with Gaussian noise and dense layers with ReLU activation. The final layer employs a sigmoid activation function for binary classification.

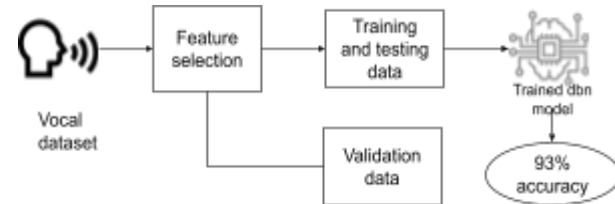


Fig. 5. Deep Belief Network Unimodal

A) Energy Function

The Energy function $E(v, h)$ of an RBM with visible units v and hidden units h is defined as:

$$E(v, h) = - \sum_i \sum_j v_i h_j w_{ij} - \sum_i b_i v_i - \sum_j c_j h_j$$

where:

- v_i is the state of visible unit i .
- h_j is the state of hidden unit j .
- w_{ij} is the weight between visible unit i and hidden unit j .
- b_i and c_j are the biases of visible unit i and hidden unit j , respectively.

B) Probability Distribution

The joint probability distribution over the visible and hidden units is defined using the energy function as:

$$P(v, h) = \frac{e^{-E(v, h)}}{Z}$$

where:

- Z is the partition function, a normalization constant ensuring the probabilities sum to 1 over all possible states.

C) Conditional Probabilities

The conditional probabilities for the states of the hidden and visible units are given by:

$$P(h_j = 1|v) = \sigma(c_j + \sum_i v_i w_{ij})$$

$$P(v_i = 1|h) = \sigma(b_i + \sum_j h_j w_{ij})$$

where:

- $\sigma(x)$ is the sigmoid function

D) Training Rule

The training of an RBM involves adjusting the weights and biases to maximize the likelihood of the training data. The update rule for the weights is given by:

$$\Delta W_{ij} = \eta (< v_i h_j >_{data} - < v_i h_j >_{model})$$

where:

- $< v_i h_j >_{data}$ is the expectation of $v_i h_j$ over the training data.
- $< v_i h_j >_{model}$ is the expectation of $v_i h_j$ over the model distribution.
- η is the learning rate.

C) Data Loading and Preprocessing

The dataset is loaded from the specified path containing voice recordings and file names. File names are read from csv file, and the data is concatenated into a comprehensive Pandas DataFrame. Min-Max scaling is applied to normalize the dataset, ensuring consistent feature scales.

We used the wrapper method for vocal feature selection.

For spiral data we applied Time series normalization. The purpose of this code appears to be time series normalization, specifically stretching each time series to a uniform length (desired_length). This can be useful for ensuring consistency in the length of time series data, which is important for certain machine learning models that expect fixed-size inputs.

D) Ensemble Formulation

We used the average probability class to combine the prediction of both the models and provide the final output.



Fig. 6. Ensemble model

E) Evaluation Metrics

Accuracy:

$$\text{Accuracy} = \frac{\text{Number Of Correct Predictions}}{\text{Total Number of Predictions}}$$

EXPERIMENTS AND RESULTS

In the experimental phase, our investigation encompassed the application of both Capsule Neural Network (CNN) and Deep Belief Network (DBN) models, resulting in individual accuracies of 70% and 93%, respectively. Furthermore, the ensemble model, amalgamating the strengths of both architectures, exhibited a commendable accuracy of 70%. These outcomes underscore the distinctive efficacy of the DBN model, while the ensemble approach leveraged the complementary attributes of both CNN and DBN for a robust overall performance.

CONCLUSION

In conclusion, the research delved into the exploration of Parkinson's disease vocal datasets, comprising 6024 instances, with a balanced distribution of 3012 samples for both Parkinson and non-Parkinson cases. Employing a wrapper method, we selected a subset of 10 attributes from the initial 23, optimizing their contribution to machine learning. The dataset, sourced from biomedical voice measurements, aimed to discern individuals with Parkinson's disease from healthy subjects based on the 'status' column.

The study further extended to the implementation of diverse machine learning models, including a Capsule Neural Network (CNN) and a Deep Belief Network (DBN). Notably, the CNN and DBN models achieved accuracies of 83% and 93%, respectively, underscoring the efficacy of the DBN architecture. Additionally, the ensemble model, combining the strengths of both CNN and DBN, exhibited a commendable accuracy of 80%.

These findings collectively highlight the significance of tailored model selection and ensemble strategies in achieving robust outcomes in medical diagnosis tasks, emphasizing the potential of deep learning techniques in the domain of vocal biomarker analysis for Parkinson's disease detection.

REFERENCES

- [1] S. Aichi, H. -C. Kim, K. younga, K. L. Hui, A. A. Al-Absi and M. Sain, "A Supervised Machine Learning Approach using Different Feature Selection Techniques on Voice Datasets for Prediction of Parkinson's Disease," 2019 21st International Conference on Advanced Communication Technology (ICACT), PyeongChang, Korea (South), 2019, pp. 1116-1121, doi: 10.23919/ICACT.2019.8701961..
- [2] K. M. M. Rao, M. S. N. Reddy, V. R. Teja, P. Krishnan, D.J. Aravindhar and M. Sambath, "Parkinson's Disease Detection Using Voice and Spiral Drawing Dataset," 2020 Third International Conference on Smart Systems and Inventive Technology (ICSSIT), Tirunelveli, India, 2020, pp. 787-791, doi: 10.1109/ICSSIT48917.2020.9214276.

- [3] H. N. Pham et al., "Multimodal Detection of Parkinson Disease based on Vocal and Improved Spiral Test," 2019 International Conference on System Science and Engineering (ICSSE), Dong Hoi, Vietnam, 2019, pp. 279-284, doi: 10.1109/ICSSE.2019.8823309.
- [4] M. Hireš, M. Gazda, L. Vavrek and P. Drotár, "Voice-Specific Augmentations for Parkinson's Disease Detection Using Deep Convolutional Neural Network," 2022 IEEE 20th Jubilee World Symposium on Applied Machine Intelligence and Informatics (SAMI), Poprad, Slovakia, 2022, pp. 000213-000218, doi: 10.1109/SAMI54271.2022.9780856.
- [5] C. R. Pereira et al., "A Step Towards the Automated Diagnosis of Parkinson's Disease: Analyzing Handwriting Movements," 2015 IEEE 28th International Symposium on Computer-Based Medical Systems, Sao Carlos, Brazil, 2015, pp. 171-176, doi: 10.1109/CBMS.2015.34.
- [6] M. Wodzinski, A. Skalski, D. Hemmerling, J. R. Orozco-Arroyave and E. Nöth, "Deep Learning Approach to Parkinson's Disease Detection Using Voice Recordings and Convolutional Neural Network Dedicated to Image Classification," 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Berlin, Germany, 2019, pp. 717-720, doi: 10.1109/EMBC.2019.8856972.
- [7] K. S., A. P. and S. R., "Parkinson Disease Detection Using Various Machine Learning Algorithms," 2022 International Conference on Advanced Computing Technologies and Applications (ICACTA), Coimbatore, India, 2022, pp.1-6, doi: 10.1109/ICACTA54488.2022.9752925.
- [8] S. Chakraborty, S. Aich, Jong-Seong-Sim, E. Han, J. Park and H. -C. Kim, "Parkinson's Disease Detection from Spiral and Wave Drawings using Convolutional Neural Networks: A Multistage Classifier Approach," 2020 22nd International Conference on Advanced Communication Technology (ICACT), Phoenix Park, Korea (South), 2020, pp. 298-303, doi: 10.23919/ICACT48636.2020.9061497.
- [9] A. Ouhmida, O. Terrada, A. Raihani, B. Cherradi and S. Hamida, "Voice-Based Deep Learning Medical Diagnosis System for Parkinson's Disease Prediction," 2021 International Congress of Advanced Technology and Engineering (ICOTEN), Taiz, Yemen, 2021, pp. 1-5, doi: 10.1109/ICOTEN52080.2021.9493456.
- [10] J. Gandhi, A. Gadekar, T. Rajabally, P. Vinayakray-Jani and D. Ambawade, "Detection of Parkinson's Disease Via a Multi-Modal Approach," 2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kharagpur, India, 2021, pp. 1-7, doi: 10.1109/ICCCNT51525.2021.9580135.
- [11] S. Raval, R. Balar and V. Patel, "A Comparative Study of Early Detection of Parkinson's Disease using Machine Learning Techniques," 2020 4th International Conference on Trends in Electronics and Informatics (ICOEI)(48184), Tirunelveli, India, 2020, pp. 509-516, doi: 10.1109/ICOEI48184.2020.9142956.
- [12] T. P. Rani, K. S. B and T. G. Prasanth, "A Voice and Spiral Drawing Dataset for Parkinson's Disease Detection and Mobile Application," 2022 International Conference on Communication, Computing and Internet of Things (IC3IoT), Chennai, India, 2022, pp. 1-5, doi: 10.1109/IC3IoT53935.2022.9767968.